

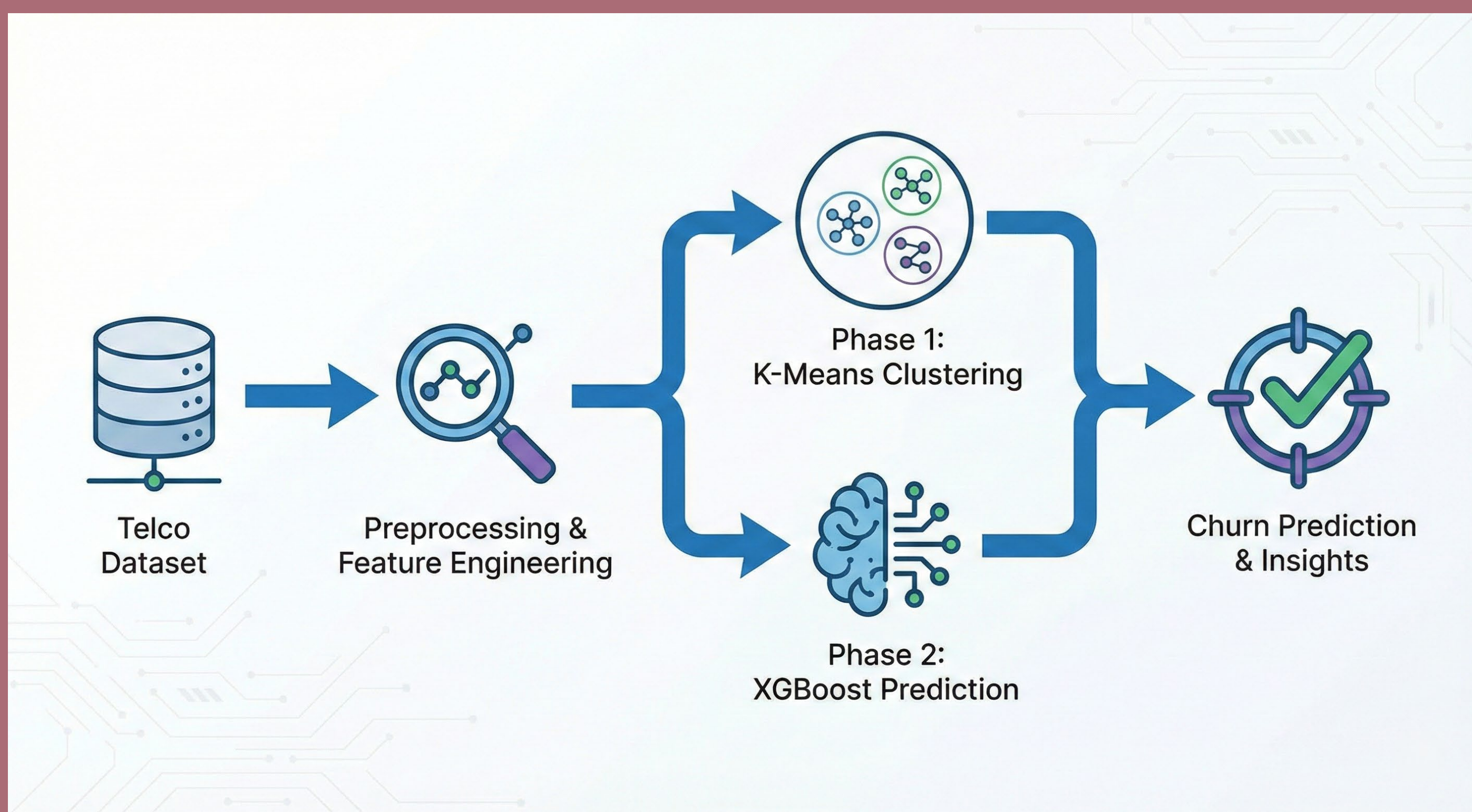
Intelligent Customer Churn Prediction in the Telecom Industry

A Hybrid Approach Using K-Means Clustering and XGBoost

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Abstract

Customer churn presents a significant financial challenge in the telecommunications industry. This project develops a proactive, data-driven system to predict churn using a hybrid approach on the Telco dataset. The methodology integrates unsupervised learning (**K-Means Clustering**) to identify behavioral segments and supervised learning (**XGBoost**) for accurate prediction. To address data imbalance, class weighting and threshold tuning were applied. The final model achieved an **accuracy of 78.61%**, identifying "Contract Type" and "Fiber Optic Service" as primary churn drivers. Furthermore, the derived "Cluster" feature ranked among the top predictors, validating the effectiveness of this hybrid strategy for targeted customer retention.



Introduction

- The Challenge:** The telecommunications industry is highly competitive and saturated, making it easy for customers to switch providers.
- The Cost:** Research shows that **Customer Acquisition Cost (CAC)** is **5 to 25 times higher** than the cost of retaining an existing customer. Therefore, even a small reduction in churn can significantly increase profitability.
- The Problem:** Traditional marketing approaches are often "reactive", acting only after the customer has decided to leave.
- Our Solution:** This project aims to transition to a "proactive" data-driven approach. We utilized the **Telco Customer Churn dataset** to develop a **Hybrid System** (K-Means Clustering + XGBoost) to identify churn drivers and predict churn accurately.

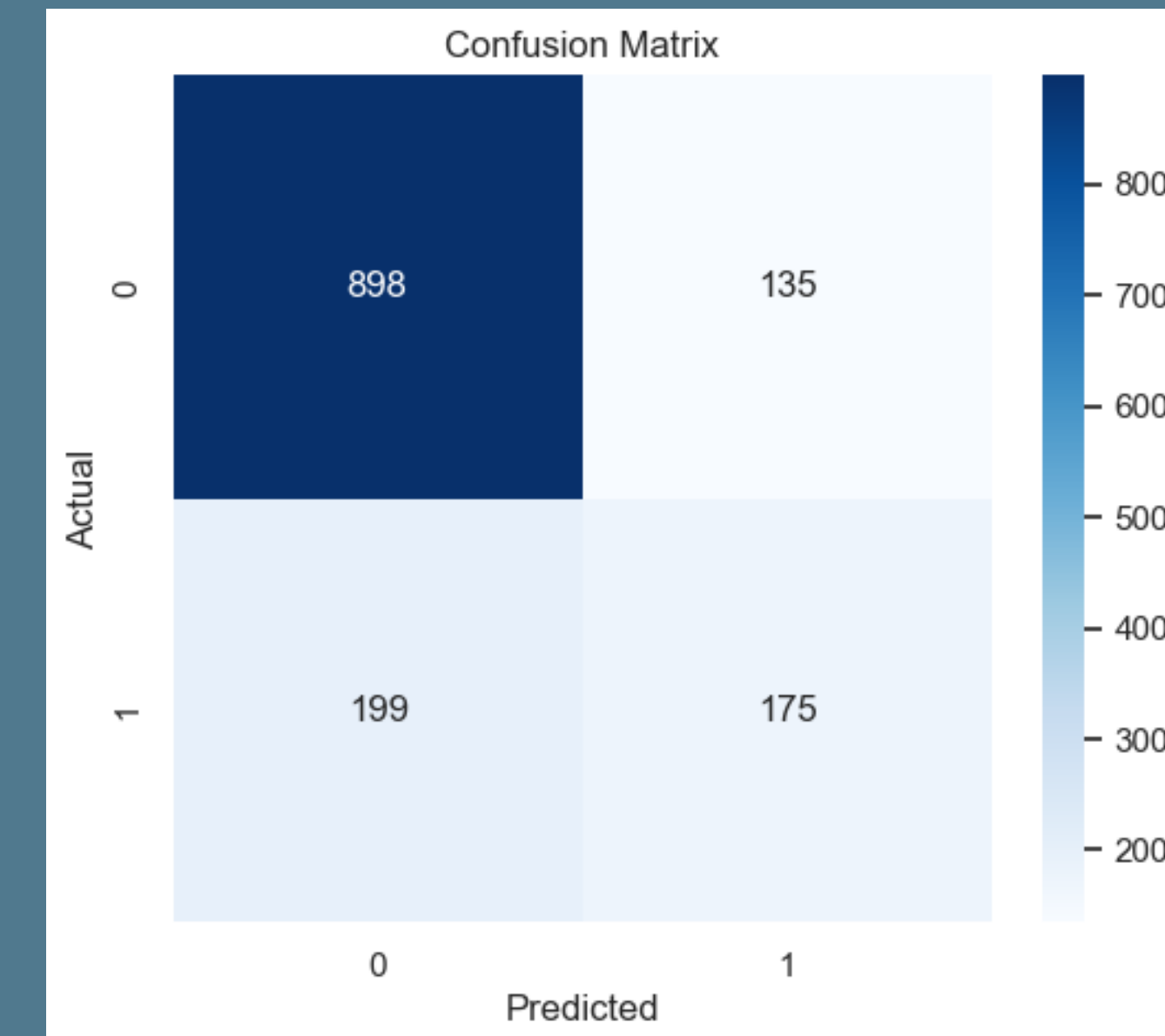
Conclusion

- Proactive Strategy:** The project successfully demonstrated the shift from reactive marketing to a **data-driven proactive approach**, enabling targeted retention efforts.
- Hybrid Success:** Integrating **K-Means Clustering** with **XGBoost** proved effective, with the derived "Cluster" feature ranking among the top 5 predictors, validating that customer segmentation enhances prediction accuracy.
- Robust Performance:** By addressing data imbalance through **Class Weighting** and **Threshold Tuning**, the model achieved a **78.61% accuracy** with a balanced Recall rate, minimizing the risk of missing churners.
- Actionable Insights:** The model "opened the black box" to reveal that **Month-to-month contracts** and **Fiber Optic service issues** are the primary root causes of churn, guiding specific retention strategies.
- Future Scope:** Future improvements could include analyzing unstructured text data (e.g., customer complaints) using **NLP** and exploring **Deep Learning** models for larger datasets.

Results

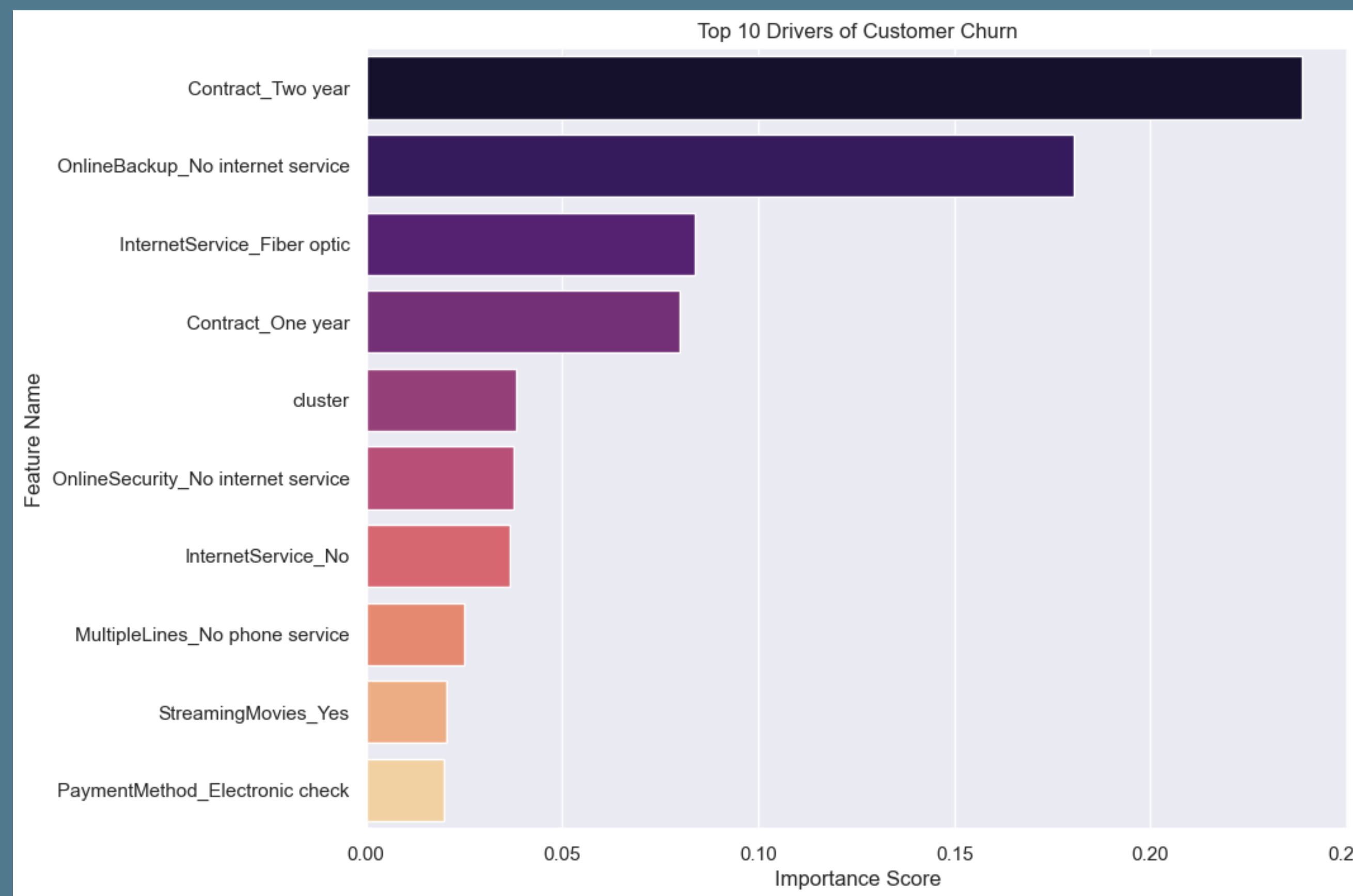
1. Model Performance

- Best Model:** **XGBoost** outperformed other algorithms (Random Forest, SVM, etc)
- Final Accuracy:** Achieved **78.61%** accuracy on the test set.
- Optimization:** Applied **Class Weighting** to handle imbalance and tuned the decision threshold to **0.79** to maximize precision



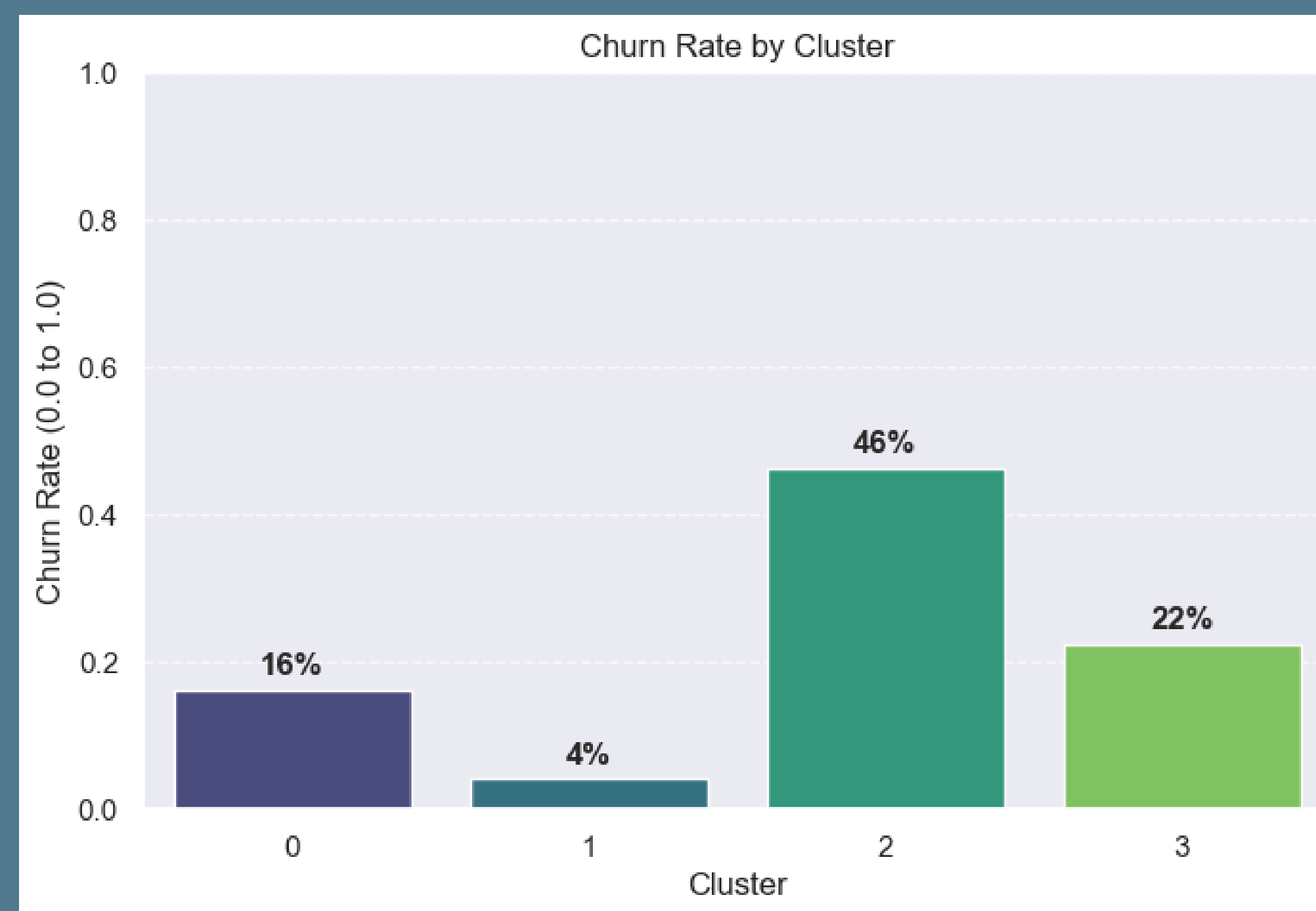
2. Key Drivers of Churn (Feature Importance)

- Contract Type:** Short-term (Month-to-month) contracts are the #1 predictor of churn.
- Internet Service:** Customers with **Fiber Optic** service show significantly higher dissatisfaction.
- Hybrid Success:** The derived "**Cluster**" feature ranked as the **5th most important predictor**, validating the effectiveness of the unsupervised learning phase.



3. Customer Segmentation Insights

- Cluster 2 High Risk:** Identified a specific segment with a **46% churn rate**.
- Cluster 1 Loyal:** Identified a low-risk segment with only **4% churn**, ideal for analyzing loyalty patterns.



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